

# Laplace and Transport equation

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## 1 Introduction

During my reading of Evans' textbook on partial differential equations, I observed that Chapter 2 introduces the fundamental solutions of several important linear equations, but often leaves the detailed derivations of these formulas to the reader. In my previous undergraduate PDE course, the discussion was primarily centered on two-dimensional bounded domains, so the  $n$ -dimensional perspective was still new to me. Since explicit solution formulas are relatively rare in the theory of partial differential equations, I found it worthwhile to work through these derivations carefully. This report therefore focuses on deriving the solution formulas for the transport equation and Laplace's equation, with particular attention to the ideas behind the  $n$ -dimensional case.

This report is organized into three main parts:

1. **The transport equation.**
2. **The non-homogeneous transport equation.**
3. **Laplace's equation.**

## 2 The Transport Equation

We first solve the homogeneous transport equation

$$\begin{cases} u_t + c \cdot Du = 0, \\ u(x, 0) = g(x). \end{cases}$$

Here  $c$  is a constant vector, and  $Du$  denotes the gradient of  $u$  with respect to the spatial variable  $x$ .

Since

$$\frac{d}{dt}u(x(t), t) = Du(x(t), t) \cdot \frac{dx}{dt} + \frac{\partial u}{\partial t}(x(t), t),$$

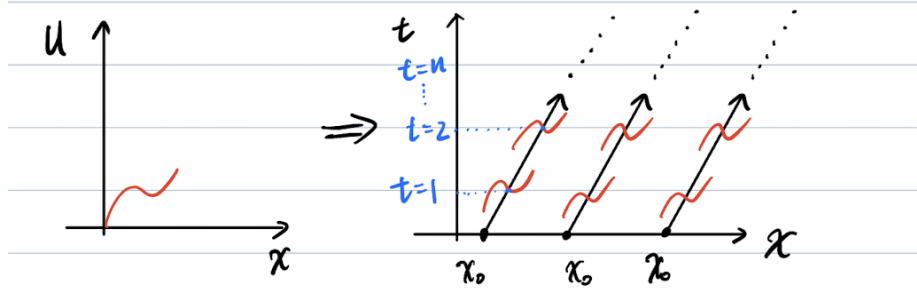


Figure 1: visualizing characteristic curve

we compare this with the PDE

$$u_t + c \cdot Du = 0.$$

Therefore, along a characteristic curve, we choose

$$\frac{dx}{dt} = c.$$

Then

$$\frac{d}{dt}u(x(t), t) = Du(x(t), t) \cdot c + u_t(x(t), t) = 0.$$

There are two important meanings here.

1. The equation

$$\frac{dx}{dt} = c$$

means that the spatial point  $x$  changes with respect to  $t$  at a constant velocity  $c$ .

2. The equation

$$\frac{d}{dt}u(x(t), t) = 0$$

means that  $u$  is not changing with respect to  $t$  along the characteristic curve.

This is very different from saying

$$\frac{\partial u}{\partial t} = 0.$$

The expression  $\frac{\partial u}{\partial t} = 0$  only measures the change of  $u$  with respect to  $t$  while keeping  $x$  fixed. However, here  $x = x(t)$  is also moving. Therefore,  $u$  and  $t$  do not have a simple one-variable relationship along the characteristic.

Now we solve

$$\frac{dx}{dt} = c.$$

Integrating both sides gives

$$x(t) = ct + k,$$

where  $k$  is a constant vector.

We want to know what  $k$  is when  $t = 0$ . Since

$$x(0) = k,$$

we define

$$x(0) := x_0.$$

Therefore,

$$k = x_0.$$

So the characteristic curve is

$$x(t) = ct + x_0.$$

Equivalently,

$$x_0 = x(t) - ct.$$

At  $t = 0$ , the initial value condition says

$$u(x(0), 0) = g(x_0).$$

Since  $u$  is constant along each characteristic curve, we get

$$u(x(t), t) = g(x_0).$$

Using

$$x_0 = x(t) - ct,$$

we obtain

$$u(x(t), t) = g(x(t) - ct).$$

Therefore, if we write the endpoint  $x(t)$  simply as  $x$ , the solution is

$$\boxed{u(x, t) = g(x - ct).}$$

### 3 The Non-Homogeneous Transport Equation

Now we solve the non-homogeneous transport equation

$$\begin{cases} u_t + b \cdot Du = f, & x \in \mathbb{R}^n, t > 0, \\ u = g, & t = 0. \end{cases}$$

Here  $b$  is a constant vector in  $\mathbb{R}^n$ , and  $Du$  denotes the gradient of  $u$  with respect to the spatial variable  $x$ .

As before, by the chain rule,

$$\frac{d}{dt}u(x(t), t) = Du(x(t), t) \cdot \frac{dx}{dt} + \frac{\partial u}{\partial t}(x(t), t).$$

Compare this with the coefficient in the PDE. Since the PDE is

$$u_t + b \cdot Du = f,$$

we choose the characteristic curve by setting

$$\frac{dx}{dt} = b.$$

Then

$$\frac{d}{dt}u(x(t), t) = Du(x(t), t) \cdot b + u_t(x(t), t) = f(x(t), t).$$

Now solve

$$\frac{dx}{dt} = b.$$

Integrating both sides gives

$$x(t) = bt + x_0.$$

Hence

$$x_0 = x(t) - bt.$$

At  $t = 0$ , the initial condition gives

$$u(x(0), 0) = g(x_0).$$

Since

$$x_0 = x(t) - bt,$$

we have

$$g(x_0) = g(x(t) - bt).$$

Now fix the final point  $(x, t)$ . Define

$$z(s) = u(x + sb, t + s).$$

The reason for this choice is that when  $s = 0$ , we get

$$z(0) = u(x, t).$$

When  $s = -t$ , we get

$$z(-t) = u(x - tb, 0) = g(x - tb).$$

Next, we compute  $z'(s)$ . By the chain rule,

$$z'(s) = \frac{d}{ds}u(x + sb, t + s).$$

Therefore,

$$z'(s) = Du(x + sb, t + s) \cdot b + u_t(x + sb, t + s).$$

By the PDE

$$u_t + b \cdot Du = f,$$

we get

$$z'(s) = f(x + sb, t + s)$$

for any point  $(x + sb, t + s)$ .

Now integrate both sides from  $-t$  to 0:

$$\int_{-t}^0 z'(s) ds = z(0) - z(-t).$$

Thus,

$$z(0) - z(-t) = u(x, t) - g(x - bt).$$

On the other hand,

$$\int_{-t}^0 z'(s) ds = \int_{-t}^0 f(x + sb, t + s) ds.$$

Therefore,

$$u(x, t) - g(x - bt) = \int_{-t}^0 f(x + sb, t + s) ds.$$

Hence the solution is

$$u(x, t) = g(x - bt) + \int_{-t}^0 f(x + sb, t + s) ds.$$

use substitution to get the exact same result as in Evans's book

## 4 Laplace's Equation

### 4.1 Reduction to a Radial ODE

Recall that the Laplace operator is defined by

$$\Delta := \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}.$$

We want to solve Laplace's equation

$$\Delta u = 0.$$

There are not too many explicit solution formulas in the full space  $\mathbb{R}^n$ . Therefore, we first suppose that the solution is radial. That is, we assume

$$u(x) = v(r),$$

where

$$r = \|x\|_2 = \sqrt{x_1^2 + \cdots + x_n^2} = (x_1^2 + \cdots + x_n^2)^{1/2}.$$

Since  $u(x) = v(r)$ , by the chain rule,

$$u_{x_i} = v'(r) \frac{\partial r}{\partial x_i}.$$

Now compute  $\partial r / \partial x_i$ . Since

$$r = (x_1^2 + \cdots + x_n^2)^{1/2},$$

we have

$$\frac{\partial r}{\partial x_i} = \frac{1}{2} (x_1^2 + \cdots + x_n^2)^{-1/2} \cdot 2x_i.$$

Therefore,

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{\sqrt{x_1^2 + \cdots + x_n^2}} = \frac{x_i}{r}.$$

Hence

$$u_{x_i} = v'(r) \frac{x_i}{r}.$$

Next, we compute  $u_{x_i x_i}$ . We have

$$u_{x_i x_i} = \frac{\partial}{\partial x_i} \left( v'(r) \frac{x_i}{r} \right).$$

By the product rule,

$$u_{x_i x_i} = v''(r) \frac{\partial r}{\partial x_i} \frac{x_i}{r} + v'(r) \frac{\partial}{\partial x_i} \left( \frac{x_i}{r} \right).$$

Since

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r},$$

the first term becomes

$$v''(r) \frac{x_i}{r} \frac{x_i}{r} = v''(r) \frac{x_i^2}{r^2}.$$

Now we compute

$$\frac{\partial}{\partial x_i} \left( \frac{x_i}{r} \right).$$

Using the quotient rule,

$$\frac{\partial}{\partial x_i} \left( \frac{x_i}{r} \right) = \frac{r \cdot 1 - x_i \frac{\partial r}{\partial x_i}}{r^2}.$$

Since

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r},$$

we get

$$\frac{\partial}{\partial x_i} \left( \frac{x_i}{r} \right) = \frac{r - \frac{x_i^2}{r}}{r^2} = \frac{r^2 - x_i^2}{r^3} = \frac{1}{r} - \frac{x_i^2}{r^3}.$$

Plugging this back into the formula for  $u_{x_i x_i}$ , we obtain

$$u_{x_i x_i} = v''(r) \frac{x_i^2}{r^2} + v'(r) \left( \frac{1}{r} - \frac{x_i^2}{r^3} \right).$$

Therefore,

$$\Delta u = \sum_{i=1}^n u_{x_i x_i} = \sum_{i=1}^n \left[ v''(r) \frac{x_i^2}{r^2} + v'(r) \left( \frac{1}{r} - \frac{x_i^2}{r^3} \right) \right].$$

Now split the summation:

$$\Delta u = \sum_{i=1}^n v''(r) \frac{x_i^2}{r^2} + \sum_{i=1}^n v'(r) \left( \frac{1}{r} - \frac{x_i^2}{r^3} \right).$$

Recall that

$$r^2 = \sum_{i=1}^n x_i^2.$$

Hence

$$\sum_{i=1}^n \frac{x_i^2}{r^2} = \frac{1}{r^2} \sum_{i=1}^n x_i^2 = \frac{r^2}{r^2} = 1.$$

Also,

$$\sum_{i=1}^n \frac{1}{r} = \frac{n}{r},$$

and

$$\sum_{i=1}^n \frac{x_i^2}{r^3} = \frac{1}{r^3} \sum_{i=1}^n x_i^2 = \frac{r^2}{r^3} = \frac{1}{r}.$$

Therefore,

$$\Delta u = v''(r) + v'(r) \left( \frac{n}{r} - \frac{1}{r} \right).$$

Thus,

$$\boxed{\Delta u = v''(r) + \frac{n-1}{r} v'(r).}$$

Since we want  $\Delta u = 0$ , the radial function  $v(r)$  must satisfy

$$\boxed{v''(r) + \frac{n-1}{r} v'(r) = 0.}$$

## 4.2 Solving the Radial ODE

Now we know that

$$\Delta u = 0$$

is equivalent to

$$v''(r) + \frac{n-1}{r}v'(r) = 0.$$

Let

$$k(r) = v'(r).$$

Then the equation becomes

$$k'(r) + \frac{n-1}{r}k(r) = 0.$$

So

$$k'(r) = -\frac{n-1}{r}k(r).$$

Hence

$$\frac{k'(r)}{k(r)} = -\frac{n-1}{r}.$$

Integrating both sides gives

$$\ln |k(r)| = -(n-1) \ln r + A.$$

Therefore,

$$e^{\ln |k(r)|} = e^{-(n-1) \ln r} \cdot e^A.$$

So

$$k(r) = Ar^{1-n}.$$

Since  $k(r) = v'(r)$ , we have

$$v'(r) = Ar^{1-n}.$$

Now solve  $v'(r) = Ar^{1-n}$ .

If  $n = 2$ , then

$$v'(r) = Ar^{-1}.$$

Therefore,

$$v(r) = A \ln r + D.$$

If  $n \geq 3$ , then

$$v'(r) = Ar^{1-n}.$$

Therefore,

$$v(r) = A \int r^{1-n} dr.$$

Since

$$\int r^{1-n} dr = \frac{r^{2-n}}{2-n},$$

we get

$$v(r) = \frac{A}{2-n} r^{2-n} + D.$$

Absorbing constants, we obtain

$$\Phi(x) \sim \begin{cases} A \ln |x| + D, & n = 2, \\ A|x|^{2-n} + D, & n \geq 3, \end{cases}$$

where  $A, D \in \mathbb{R}$ .

To get the same solution as Evans' book, we need to choose the correct constant by requiring

$$-\Delta \Phi = \delta_0.$$

Now we apply Green's theorem. For the ball  $B_R(0)$ ,

$$\int_{B_R(0)} -\Delta \Phi \, dx = - \int_{\partial B_R(0)} \frac{\partial \Phi}{\partial \nu} \, dS = \int_{B_R(0)} \delta_0(x) \, dx = 1. \quad (1)$$

• **Case 1**  $n = 2$

Recall that when  $n = 2$ ,

$$\Phi(x) = A \ln |x| + D.$$

Let

$$r = |x|.$$

Then

$$\frac{\partial \Phi}{\partial r} = \frac{A}{r}.$$

On  $\partial B_R(0)$ , we have  $r = R$ . In two dimensions,

$$|\partial B_R(0)| = 2\pi R.$$

Hence

$$- \int_{\partial B_R(0)} \frac{\partial \Phi}{\partial \nu} \, dS = - \int_{\partial B_R(0)} \frac{A}{r} \, dS.$$

Since  $r = R$  on  $\partial B_R(0)$ , this becomes

$$- \int_{\partial B_R(0)} \frac{A}{R} \, dS = - \frac{A}{R} |\partial B_R(0)|.$$

Using

$$|\partial B_R(0)| = 2\pi R,$$

we get

$$-\frac{A}{R} \cdot 2\pi R = -2\pi A.$$

Since

$$-\Delta\Phi = \delta_0,$$

we need

$$-2\pi A = 1.$$

Therefore,

$$A = -\frac{1}{2\pi}.$$

Thus, for  $n = 2$ ,

$$\boxed{\Phi(x) = -\frac{1}{2\pi} \ln |x|}.$$

• **Case 2**  $n \geq 3$

Recall that when  $n \geq 3$ ,

$$\Phi(x) = B|x|^{2-n}.$$

Let

$$r = |x|.$$

Then

$$\Phi(r) = Br^{2-n}.$$

Differentiate with respect to  $r$ :

$$\frac{\partial\Phi}{\partial r} = B(2-n)r^{1-n}.$$

Equivalently,

$$\frac{\partial\Phi}{\partial r} = -B(n-2)r^{1-n}.$$

So

$$-\int_{\partial B_R(0)} \frac{\partial\Phi}{\partial\nu} dS = -\int_{\partial B_R(0)} [-B(n-2)R^{1-n}] dS.$$

Therefore,

$$-\int_{\partial B_R(0)} \frac{\partial\Phi}{\partial\nu} dS = B(n-2)R^{1-n}|\partial B_R(0)|.$$

In  $\mathbb{R}^n$ , we know that

$$|\partial B_R(0)| = n\alpha(n)R^{n-1},$$

where Evans define it  $\alpha(n)$  the volume of the unit ball in  $\mathbb{R}^n$ .

Thus,

$$B(n-2)R^{1-n}|\partial B_R(0)| = B(n-2)R^{1-n} \cdot n\alpha(n)R^{n-1}.$$

Since

$$R^{1-n}R^{n-1} = 1,$$

we get

$$B(n-2)n\alpha(n).$$

Since (1), we have

$$B(n-2)n\alpha(n) = 1.$$

Therefore,

$$B = \frac{1}{n(n-2)\alpha(n)}.$$

Thus, for  $n \geq 3$ ,

$$\Phi(x) = \frac{|x|^{2-n}}{n(n-2)\alpha(n)}$$

Therefore, the final form of the fundamental solution is

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi} \ln |x|, & n = 2, \\ \frac{1}{n(n-2)\alpha(n)} |x|^{2-n}, & n \geq 3. \end{cases}$$

## 5 Conclusion

In working through the derivations of the transport equation and Laplace's equation, one overarching idea stands out: both problems are solved by reducing a partial differential equation to a collection of ordinary differential equations. For the transport equation, this reduction is achieved through the method of characteristics, where we identify special curves along which the PDE simplifies to a total derivative. For Laplace's equation, Evans exploits radial symmetry to reduce the full  $n$ -dimensional Laplacian to a single ODE in the variable  $r$ . Although the two techniques look different on the surface, they share the same underlying idea which is find the right structure in the problem, and use it to collapse the complexity of a PDE into something more manageable.

Reading through Evans also raised some questions that I hope to explore further. One concerns the relationship between point sources and the Dirac delta function. Evans defines the fundamental solution via  $-\Delta\Phi = \delta_0$ , which was initially puzzling: what does it mean to say a differential operator applied to a function equals something that is not a function at all? I came to understand  $\delta_0$  as an idealized point source, like an object that concentrates all its mass at a single point, but making this precise requires the language of distributions, which goes beyond what is covered here. I look forward to understanding this more rigorously in future coursework.